Closed-Form Solutions for Improving Sensor Position Accuracy via Inter-Sensor Ranges

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ABSTRACT

In this paper, we investigate the problem of improving erroneous sensor positions using inter-sensor range measurements. The existing multidimentional scaling (MDS)-based two-step algorithm performs poorly in three key scenarios: (i) incomplete connectivity of range measurements; (ii) moderate or severe range measurement noise; and (iii) three-dimensional settings. To overcome these limitations, we develop two novel closed-form solutions. The first, an error compensation (EC) method, applies the first-order Taylor series expansion to inter-sensor range measurements and formulates a linear weighted least-squares (WLS) estimator for sensor position error. The corrected sensor positions are then obtained by subtracting the estimated error from the noisy measurements. The second, an approximated maximum likelihood (AML) method, replaces the nonconvex component of the maximum likelihood cost function with a quadratic form, resulting in a new WLS estimator for sensor positions. Simulation results show that both proposed algorithms outperform the MDS-based algorithm, with the AML achieving the best performance in moderate to severe range measurement noise.

***Index Terms*—** Erroneous sensor positions, error compensation, approximated maximum likelihood, weighted least-squares

1. INTRODUCTION

Wireless sensor networks (WSNs) have broad applications such as battlefield surveillance, intruder detection, health monitoring, environmental monitoring, transport logistics, and smart cities [1, 2, 3], many of which depend on position information. Precise knowledge of sensor positions is essential for source localization in WSNs [4, 5, 6, 7, 8] since the accuracy of location estimates is highly sensitive to sensor position uncertainties [9, 10]. In practice, sensor position errors are inevitable, arising, for example, from sensor drifts caused by sea waves or from inaccuracies in self-localization procedures [11].

Sun *et al.* [11] proposed a two-step method leverages accurate inter-sensor range measurements to refine erroneous sensor positions. In the first step, fully connected inter-sensor range measurements are used to form a multidimensional scaling (MDS)-based estimator of sensor position, yielding a shifted, rotated and possibly reflected version of the true sensor configuration. In the second step, the true positions of the sensors are recovered by formulating and solving a quadratic optimization problem with a single quadratic constraint, which can be recast into a generalized trust region sub-problem (GTRS) and solved globally via bisection searching. However, the algorithm in [11] fails under three common scenarios: (i) incomplete connectivity of range measurements, since the MDS step requires all inter-sensor ranges to be available; (ii) moderate or severe measurement noise, as the method assumes highly accurate ranges; and (iii) three-dimensional (3D) deployment, since the GTRS formulation is restricted to 2D cases and its extension to 3D is neither addressed nor straightforward. Recently, Zou *et al.* [12] developed an efficient weighted least-squares (WLS) algorithm for the problem of improving noisy sensor positions using noisy inter-sensor AOA measurements.

In this paper, we address the problem of refining erroneous sensor positions using inter-sensor range measurements. We develop two closed-form solutions, weighted least-squares (WLS) based solutions that not only overcome the three limitations of the algorithm in [11], but also reduce the computational complexity. The main contributions of this work are the development of two closed-form solutions, namely the error compensation (EC) method and the approximate maximum likelihood (AML) method, and the demonstration, through simulations, that both effectively overcome the limitations of the existing MDS-based approach. In particular, the AML method exhibits superior performance under moderate and severe measurement noise.

The remainder of this paper is organized as follows. Section II formulates the problem. Section III analyzes the Cramér-Rao lower bounds (CRLB). Section IV details the proposed methods. Section V presents simulation results to validate their performance. Finally, Section VI concludes this paper.

2. PROBLEM STATEMENT

Consider a wireless sensor network (WSN) with M sensor nodes in a 3D space. The known but inaccurate sensor positions are expressed as

$$\mathbf{s}_i = \mathbf{s}_i^0 + \Delta\mathbf{s}_i,\ i = 1, \cdots, M \tag{1}$$

where $\mathbf{s}_i^0 = [x_i^0, y_i^0, z_i^0]^T$ denotes the unknown true sensor position, and $\Delta\mathbf{s}_i$ is the sensor position error. The sensor position error vector $\Delta\mathbf{s} = [\Delta\mathbf{s}_1^T, \cdots, \Delta\mathbf{s}_M^T]^T \in \mathbb{R}^{3M\times 1}$ is modeled as a zero mean Gaussian random vector with covariance matrix $\mathbf{Q}_s$ [9].

The measured inter-sensor range between the i-th sensor and the j-th sensor can be expressed as

$$r_{ij} = \|\mathbf{s}_j^0 - \mathbf{s}_i^0\| + n_{ij},\ 1 \le i < j \le M \tag{2}$$

where noise vector $\mathbf{n} = [n_{12}, n_{13}, \cdots, n_{M-1M}]^T \in \mathbb{R}^{\frac{M(M-1)}{2}\times 1}$ is a zero mean Gaussian random vector with covariance matrix $\mathbf{Q}_n$ [10].

The next step is to use the known measurements (r_{ij}, $\mathbf{s}_i$, $\mathbf{Q}_n$, and $\mathbf{Q}_s$) to estimate the unknown parameters $\mathbf{s}_i^0$.

3. CRLB ANALYSIS

By incorporating inter-sensor range measurements, the Fisher information matrix (FIM) for the sensor positions, expressed below, is derived directly from the statistical model of these measurements, reflecting how measurement geometry and noise influence the achievable localization accuracy.

$$\mathbf{I}(\mathbf{s}^0) = \mathbf{E}^T\mathbf{Q}_n^{-1}\mathbf{E} + \mathbf{Q}_s^{-1} \tag{3}$$

where $\mathbf{E} \in \mathbb{R}^{\frac{M(M-1)}{2}\times 3M}$, $\mathbf{s}^0 = [\mathbf{s}_1^{0T}, \cdots, \mathbf{s}_M^{0\,T}]^T \in \mathbb{R}^{3M\times 1}$, and

$$[\mathbf{E}]_{m,3i-2:3i} = \frac{\mathbf{s}_i^0 - \mathbf{s}_j^0}{\left\|\mathbf{s}_j^0 - \mathbf{s}_i^0\right\|}, \tag{4a}$$

$$[\mathbf{E}]_{m,3j-2:3j} = \frac{\mathbf{s}_j^0 - \mathbf{s}_i^0}{\left\|\mathbf{s}_j^0 - \mathbf{s}_i^0\right\|}. \tag{4b}$$

Note that there exists a bidirectional mapping between i, j, and m in Eq. (4) and the undefined entries of $\mathbf{E}$ are all zeros. The mapping is:

$$m = 1 \Leftrightarrow i = 1,\ j = 2 \tag{5a}$$

$$m = 2 \Leftrightarrow i = 1,\ j = 3 \tag{5b}$$

$$\vdots$$

$$m = M-1 \Leftrightarrow i = 1,\ j = M \tag{5c}$$

$$m = M \Leftrightarrow i = 2,\ j = 3 \tag{5d}$$

$$\vdots$$

$$m = \frac{M(M-1)}{2} \Leftrightarrow i = M-1,\ j = M. \tag{5e}$$

It is straightforward to obtain

$$\mathbf{I}(\mathbf{s}^0) \succ \mathbf{Q}_s^{-1}, \tag{6}$$

where $\mathbf{A} \succ \mathbf{B}$ means $\mathbf{A} - \mathbf{B}$ is a positive definite matrix. Then

$$\mathbf{Q}_s \succ \mathbf{I}^{-1}(\mathbf{s}^0). \tag{7}$$

This inequality shows that incorporating inter-sensor range measurements reduces the CRLB for sensor positions. In other words, effectively exploiting noisy inter-sensor range measurements can enhance the accuracy of sensor position estimates [12].

Finally, the improved CRLB of $\mathbf{s}_i^0$ is written as

$$\begin{aligned}\text{CRLB}(\mathbf{s}_i^0) = \left[\mathbf{I}^{-1}(\mathbf{s}^0)\right]_{3i-2,3i-2} + \left[\mathbf{I}^{-1}(\mathbf{s}^0)\right]_{3i-1,3i-1} \\ + \left[\mathbf{I}^{-1}(\mathbf{s}^0)\right]_{3i,3i}.\end{aligned} \tag{8}$$

4. PROPOSED METHODS

4.1. Error Compensation Method

Substituting the noisy sensor positions into (2) and applying the first-order Taylor approximation to r_{ij} at $\mathbf{s}_i$ and $\mathbf{s}_j$ yield

$$\begin{aligned}r_{ij} &= \|\mathbf{s}_j - \mathbf{s}_i - \Delta\mathbf{s}_j + \Delta s_i\| + n_{ij} \\ &\approx \|\mathbf{s}_j - \mathbf{s}_i\| + \frac{(\mathbf{s}_j - \mathbf{s}_i)^T(\Delta\mathbf{s}_i - \Delta\mathbf{s}_j)}{\|\mathbf{s}_j - \mathbf{s}_i\|} + n_{ij}.\end{aligned} \tag{9}$$

Moving the norm to the left-side of the equation yields the following expression:

$$r_{ij} - \|\mathbf{s}_j - \mathbf{s}_i\| \approx \frac{(\mathbf{s}_j - \mathbf{s}_i)^T(\Delta\mathbf{s}_i - \Delta s_j)}{\|\mathbf{s}_j - \mathbf{s}_i\|} + n_{ij}. \tag{10}$$

Eq. (10) can be further expressed as

$$\frac{(\mathbf{s}_j - \mathbf{s}_i)^T}{\|\mathbf{s}_j - \mathbf{s}_i\|}\Delta\mathbf{s}_i - \frac{(\mathbf{s}_j - \mathbf{s}_i)^T}{\|\mathbf{s}_j - \mathbf{s}_i\|}\Delta\mathbf{s}_j \approx r_{ij} - \|\mathbf{s}_j - \mathbf{s}_i\| - n_{ij}. \tag{11}$$

Stacking the equations in (11) yields

$$\mathbf{G}\Delta\mathbf{s} \approx \mathbf{h} - \mathbf{n} \tag{12}$$

where

$$\mathbf{G} \in \mathbb{R}^{\frac{M(M-1)}{2}\times 3M}, \mathbf{h} \in \mathbb{R}^{\frac{M(M-1)}{2}\times 1}, \tag{13a}$$

$$[\mathbf{G}]_{m,3i-2:3i} = \frac{(\mathbf{s}_j - \mathbf{s}_i)^T}{\|\mathbf{s}_j - \mathbf{s}_i\|}, \tag{13b}$$

$$[\mathbf{G}]_{m,3j-2:3j} = -\frac{(\mathbf{s}_j - \mathbf{s}_i)^T}{\|\mathbf{s}_j - \mathbf{s}_i\|}, \tag{13c}$$

$$h(m) = r_{ij} - \|\mathbf{s}_j - \mathbf{s}_i\|. \tag{13d}$$

Note again that the undefined entries of $\mathbf{G}$ are all zeros.

The WLS estimate of $\Delta\mathbf{s}$ is

$$\Delta\hat{\mathbf{s}} = (\mathbf{G}^T\mathbf{Q}_n^{-1}\mathbf{G})^\dagger\mathbf{G}^T\mathbf{Q}_n^{-1}\mathbf{h}, \tag{14}$$

where $(\cdot)^\dagger$ denotes the pseudo-inverse and the sum of each row of $\mathbf{G}$ equals zero. In other words, $\mathbf{G}\mathbf{1}_{3M,1} = \mathbf{0}_{\frac{M(M-1)}{2},1}$, where $\mathbf{1}_{3M,1}$ is a column vector of ones and $\mathbf{0}_{\frac{M(M-1)}{2},1}$ is a column vector of zeros.

Theses analyses allow us to obtain the following equation:

$$\mathbf{G}^T\mathbf{Q}_n^{-1}\mathbf{G}\mathbf{1}_{3M,1} = 0 \times \mathbf{1}_{3M,1}, \tag{15}$$

and $\mathbf{G}^T\mathbf{Q}_n^{-1}\mathbf{G}$ is singular since it has a zero eigenvalue.

The final estimate of $\mathbf{s}_i^0$ is expressed as

$$\hat{\mathbf{s}}_i^0 = \mathbf{s}_i - \Delta\hat{\mathbf{s}}(3i-2:3i),\ i = 1, \cdots, M. \tag{16}$$

4.2. Approximated Maximum Likelihood Method

Let $\mathbf{s} = [\mathbf{s}_1^T, \cdots, \mathbf{s}_M^T]^T$, $\mathbf{r} = [r_{12}, r_{13}, \cdots, r_{M-1M}]^T$, and $\mathbf{d} = [\|\mathbf{s}_2^0 - \mathbf{s}_1^0\|, \|\mathbf{s}_3^0 - \mathbf{s}_1^0\|, \cdots, \|\mathbf{s}_M^0 - \mathbf{s}_{M-1}^0\|]^T$. Since the range noise vector $\mathbf{n}$ and sensor position error vector $\Delta\mathbf{s}$ are mutually independent, the maximum likelihood estimator (MLE) of the sensor positions is formulated as

$$\min_{\mathbf{s}^0} \quad (\mathbf{r} - \mathbf{d})^T\mathbf{Q}_n^{-1}(\mathbf{r} - \mathbf{d}) + (\mathbf{s} - \mathbf{s}^0)^T\mathbf{Q}_s^{-1}(\mathbf{s} - \mathbf{s}^0). \tag{17}$$

The first part of the above cost function corresponds to the inter-sensor range measurements, which is highly nonlinear and nonconvex with respect to $\mathbf{s}^0$. The second part of the above cost function corresponds to the noisy sensor position measurements, which is quadratic with respect to $\mathbf{s}^0$.

Although the MLE is asymptotically efficient, it is highly nonlinear and nonconvex, which makes it very difficult to solve directly. Next, we develop a quadratic form to approximate the first part of the cost function.

Substituting the noisy sensor positions into (2) yields

$$r_{ij} = \|\mathbf{s}_j - \mathbf{s}_i - \Delta\mathbf{s}_j + \Delta s_i\| + n_{ij}. \tag{18}$$

Squaring both sides of Eq. (18) and ignoring the second-order terms of noise Δs_i, Δs_j, and n_{ij} yield

$$r_{ij}^2 \approx \|\mathbf{s}_j - \mathbf{s}_i\|^2 - 2(\mathbf{s}_j - \mathbf{s}_i)^T(\Delta\mathbf{s}_j - \Delta s_i) + 2\|\mathbf{s}_j - \mathbf{s}_i\|n_{ij}. \tag{19}$$

Eq. (19) can be rewritten as

$$-(\mathbf{s}_j - \mathbf{s}_i)^T \Delta \mathbf{s}_i + (\mathbf{s}_j - \mathbf{s}_i)^T \Delta \mathbf{s}_j \approx \frac{1}{2}(\|\mathbf{s}_j - \mathbf{s}_i\|^2 - r_{ij}^2) + \|\mathbf{s}_j - \mathbf{s}_i\| n_{ij}. \quad (20)$$

Stacking the equations in (20) yields the following vector-matrix form:

$$\mathbf{A}\Delta\mathbf{s} \approx \mathbf{b} - \mathbf{B}\mathbf{n} \quad (21)$$

or

$$\mathbf{A}(\mathbf{s} - \mathbf{s}^0) \approx \mathbf{b} - \mathbf{B}\mathbf{n} \quad (22)$$

where

$$\mathbf{A} \in \mathbb{R}^{\frac{M(M-1)}{2} \times 3M}, \mathbf{b} \in \mathbb{R}^{\frac{M(M-1)}{2} \times 1}, \mathbf{B} \in \mathbb{R}^{\frac{M(M-1)}{2} \times \frac{M(M-1)}{2}}, \quad (23a)$$

$$[\mathbf{A}]_{m,3i-2:3i} = -(\mathbf{s}_j - \mathbf{s}_i)^T, \quad (23b)$$

$$[\mathbf{A}]_{m,3j-2:3j} = (\mathbf{s}_j - \mathbf{s}_i)^T, \quad (23c)$$

$$b(m) = \frac{1}{2}(\|\mathbf{s}_j - \mathbf{s}_i\|^2 - r_{ij}^2), \quad (23d)$$

$$\mathbf{B} = \text{diag}([\|\mathbf{s}_2 - \mathbf{s}_1\|, \|\mathbf{s}_3 - \mathbf{s}_1\|, \cdots, \|\mathbf{s}_M - \mathbf{s}_{M-1}\|]). \quad (23e)$$

Note that the undefined entries of $\mathbf{A}$ are all zeros.

Eq. (22) allows us to formulate a quadratic optimization with respect to $\mathbf{s}^0$ as follows:

$$\min_{\mathbf{s}^0} \quad (\mathbf{A}\mathbf{s}^0 + \mathbf{b} - \mathbf{A}\mathbf{s})^T \mathbf{W} (\mathbf{A}\mathbf{s}^0 + \mathbf{b} - \mathbf{A}\mathbf{s}) \quad (24)$$

where

$$\mathbf{W} = (\text{E}[\mathbf{B}\mathbf{n}\mathbf{n}^T\mathbf{B}^T])^{-1} = (\mathbf{B}\mathbf{Q}_n\mathbf{B}^T)^{-1}. \quad (25)$$

Replacing the first part of the cost function in (17) with the quadratic form in (24) results in the AML estimate problem:

$$\min_{\mathbf{s}^0} \quad (\mathbf{A}\mathbf{s}^0 + \mathbf{b} - \mathbf{A}\mathbf{s})^T \mathbf{W} (\mathbf{A}\mathbf{s}^0 + \mathbf{b} - \mathbf{A}\mathbf{s}) + (\mathbf{s} - \mathbf{s}^0)^T \mathbf{Q}_s^{-1} (\mathbf{s} - \mathbf{s}^0). \quad (26)$$

The above problem is equivalent to

$$\min_{\mathbf{s}^0} \quad {\mathbf{s}^0}^T \mathbf{A}^T \mathbf{W} \mathbf{A} \mathbf{s}^0 + 2{\mathbf{s}^0}^T \mathbf{A}^T \mathbf{W} (\mathbf{b} - \mathbf{A}\mathbf{s}) + {\mathbf{s}^0}^T \mathbf{Q}_s^{-1} \mathbf{s}^0 - 2{\mathbf{s}^0}^T \mathbf{Q}_s^{-1} \mathbf{s}. \quad (27)$$

where two constant terms are discarded since they do not affect the optimization results.

Finally, letting the gradient of the objective function in (27) with respect to $\mathbf{s}^0$ to zero, we have

$$2\mathbf{A}^T\mathbf{W}\mathbf{A}\mathbf{s}^0 + 2\mathbf{A}^T\mathbf{W}(\mathbf{b} - \mathbf{A}\mathbf{s}) + 2\mathbf{Q}_s^{-1}\mathbf{s}^0 - 2\mathbf{Q}_s^{-1}\mathbf{s} = \mathbf{0} \quad (28)$$

which yields the estimate of sensor position expressed as

$$\hat{\mathbf{s}}^0 = (\mathbf{A}^T\mathbf{W}\mathbf{A} + \mathbf{Q}_s^{-1})^{-1} \left(\mathbf{Q}_s^{-1}\mathbf{s} + \mathbf{A}^T\mathbf{W}(\mathbf{A}\mathbf{s} - \mathbf{b})\right). \quad (29)$$

Table 1. Sensor Positions for 3D Scenarios (m)

No.	1	2	3	4
x	40	50	-50	-60
y	30	-50	70	60
z	0	50	30	80

Table 2. Sensor Positions for 2D Scenarios (m)

No.	1	2	3	4
x	0	-50	-25	25
y	0	0	43.3	-43.3

5. SIMULATION RESULTS

This section presents extensive numerical simulations to validate the performance of the two proposed algorithms, in comparison with the algorithm in [11] (label as MDS) and the CRLB. The estimate obtained from error compensation method is denoted as EC, and that from the approximate maximum likelihood method is denoted as AML.

Both 3D and 2D scenarios are considered in the simulations, with corresponding sensor positions listed in Tables 1 and 2, respectively. The sensor position errors are independent and identically distributed zero-mean Gaussian random variables with variance σ_s^2, i.e., $\mathbf{Q}_s = \sigma_s^2 \mathbf{I}_{3M}$ [9]. Similarly, range measurements noises are assumed to be independent and identically distributed zero-mean Gaussian variables with variance σ_r^2, i.e., $\mathbf{Q}_n = \sigma_r^2 \mathbf{I}_{\frac{M(M-1)}{2}}$ [10].

For clear comparison, we use the average RMSE and average CRLB of sensor positions defined as [12]

$$\text{RMSE} = \sqrt{\frac{1}{MN} \sum_{n=1}^{N} \sum_{i=1}^{M} \|\hat{\mathbf{s}}_{i,n} - \mathbf{s}_i^0\|^2}, \quad (30)$$

$$\text{CRLB} = \sqrt{\frac{1}{M} \text{Tr}(\mathbf{I}^{-1}(\mathbf{s}^0))}, \quad (31)$$

where $\mathbf{M}$ is the number of sensors, N is the number of Monte Carlo realizations ($N = 2000$), $\hat{\mathbf{s}}_{i,n}$ is the estimate of $\mathbf{s}_i^0$ in n-th realization, $\text{Tr}(\cdot)$ denotes trace of a matrix.

5.1. Three-dimensional Scenarios

Since the MDS method in [11] is limited to 2D scenarios, it is excluded from the following comparisons. The following observations can be drawn from Figs. 1 and 2:

- Both the AML and EC methods can reach the CRLB when range noise is sufficiently small (i.e., when σ_r is smaller than -10 dB-meter, or 0.1 m), and the AML method outperforms the EC method when σ_r is greater than 0.1 m;
- The EC method exhibits a threshold effect with respect to range measurement noise, arising from the use of the first-order Taylor approximation in its derivation.
- The average sensor position error can be decreased from $1.73m$ ($\sqrt{\text{Tr}(\mathbf{Q}_s)/M}$) to $1.24m$ when $\sigma_r = 0.1m$ and $\sigma_s = 1m$, and improving range accuracy does not significantly enhance the sensor position accuracy when $\sigma_r < 0.1m$.

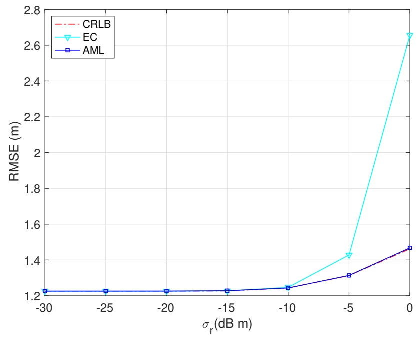

Fig. 1. RMSE versus σ_r $(\sigma_s = 1m)$ in a 3D scenario.

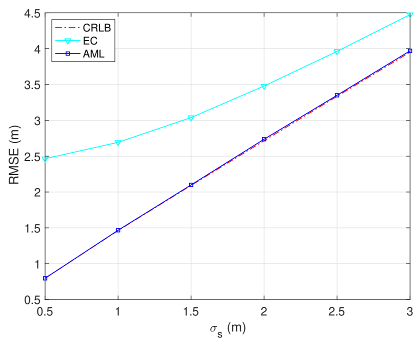

Fig. 2. RMSE versus σ_s $(\sigma_r = 1m)$ in a 3D scenario.

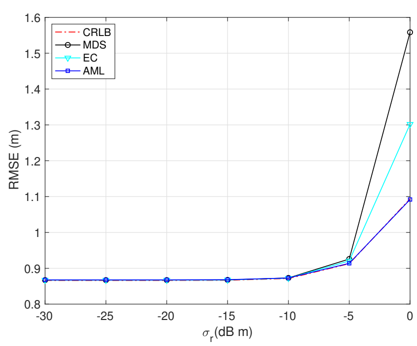

Fig. 3. RMSE versus σ_r $(\sigma_s = 1m)$ in a 2D scenario.

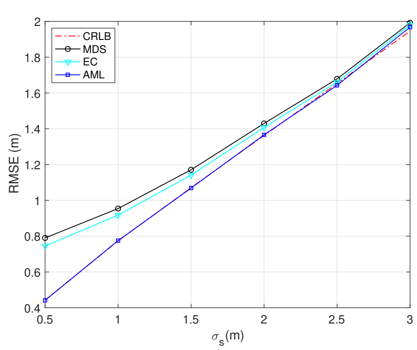

Fig. 4. RMSE versus σ_s $(\sigma_r = 1m)$ in a 2D scenario.

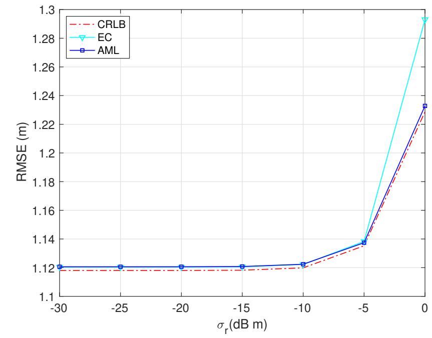

Fig. 5. RMSE versus σ_r $(\sigma_s = 1m)$ for the incomplete inter-sensor range measurements in a 2D scenario.

5.2. Two-dimensional Scenarios

Results for a 2D scenario are shown in Figs. 3 and 4. Fig. 3 shows that all three algorithms compared attain the CRLB when σ_r is smaller than 0.1 m. The EC and MDS methods deviate from the CRLB when σ_r is greater than 0.1 m, but the AML still reaches the CRLB. Fig. 4 shows that the EC performs slightly better than the MDS and the AML has the best performance.

5.3. Incomplete Connectivity of Range Measurements Scenarios

This scenario evaluates the performance of the proposed algorithms when the inter-sensor range measurements are incomplete. Such scenarios, arise, for example, when the distance between two sensors exceeds their maximum communications distance, or when links are blocked by obstacles, leading to missing range measurements. Because the MDS method cannot operate with incomplete connectivity of range measurements, it is excluded from Fig. 5, where only three range measurements are available (namely r_{12}, r_{23}, and r_{34}). A comparison between Fig. 3 and Fig. 5 shows that the incomplete range measurements results in a higher CRLB, specifically, $0.87m$ (fully connected) versus $1.12m$ (incompletely connected) when $\sigma_r = 0.1m$ and $\sigma_s = 1m$. Furthermore, Fig. 5 shows that AML consistently outperforms EC when range measurements noise is not sufficiently small.

6. CONCLUSIONS

In this paper, we investigated the problem of refining erroneous sensor positions with inter-sensor range measurements. First, a CRLB analysis was provided to demonstrate that the incorporating inter-sensor range measurements improves the achievable accuracy of sensor positions. Second, we developed two closed-form solutions, namely the EC and the AML methods. Finally, simulation results verified the robustness and effectiveness of both algorithms compared with the existing MDS-based approach, with the AML method achieving the best performance under moderate to severe measurement noise.

7. REFERENCES

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